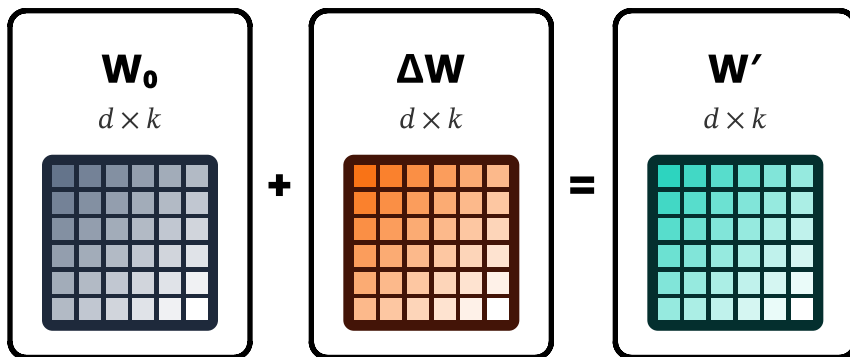
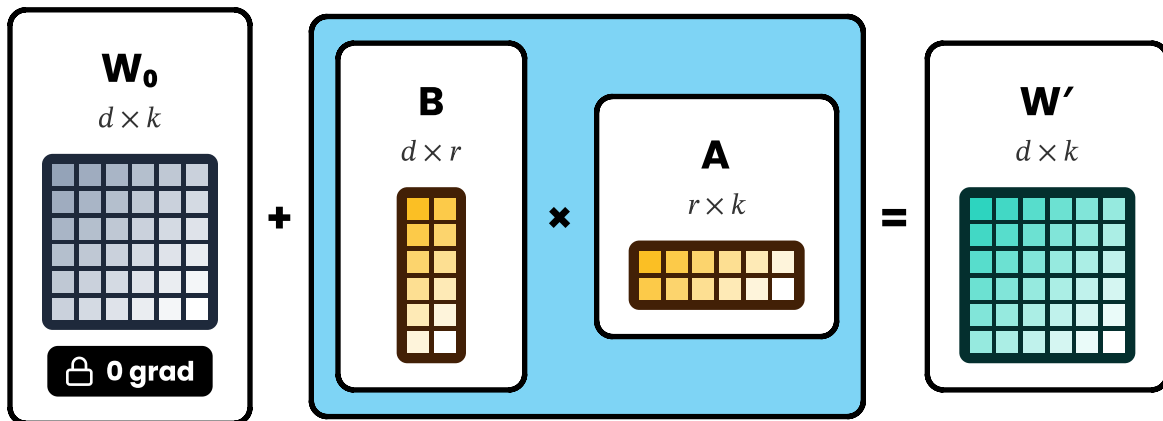


Full fine-tuning — every weight is trained



LoRA — freeze W_0 , train one low-rank branch



Trainable weights, same layer

full fine-tuning · $d \cdot k$ — every weight

LoRA · $r(d + k)$ → orders of magnitude fewer (not to scale)